Mathematics & Advanced Statistics
Practical Exam
Theory Report

1. Mean, Median and Mode

Mean: The arithmetic average of all observations.
Formula: Mean = Sum of observations / Number of observations.

Median: The middle value after arranging the data in ascending order. It is less affected by extreme values.

Mode: The value that occurs most frequently in the dataset.

Application: In the employee dataset, these measures are used to summarize the salary distribution.

2. Range and Variance

Range = Maximum Value − Minimum Value.

Variance measures how much the data values are spread around the mean. A larger variance indicates greater variability.

Application: Variance of Projects_Completed shows whether employees complete a similar number of projects or differ widely.

3. Normal Distribution vs Poisson Distribution

Normal Distribution:
- Continuous probability distribution.
- Bell-shaped and symmetric.
- Defined by mean and standard deviation.

Poisson Distribution:
- Discrete probability distribution.
- Models the number of events occurring in a fixed interval.
- Defined by the parameter λ (lambda).

4. Skewness

Skewness measures the asymmetry of a distribution.

- Skewness = 0 → Symmetric distribution
- Positive Skewness → Long right tail
- Negative Skewness → Long left tail

5. Conditional Probability

Conditional probability is the probability of an event occurring given that another event has already occurred.

Formula:
$P(A|B) = P(A \cap B) / P(B)$

Example:
Probability that an employee is promoted given that the Performance Score is greater than 80.

6. Mutually Exclusive Events

Two events are mutually exclusive if they cannot occur at the same time.

Formula:
$P(A \cup B) = P(A) + P(B)$

Example:
An employee cannot simultaneously have Promotion_Status as both Yes and No.

7. Bayes' Theorem

Bayes' Theorem updates the probability of an event based on new information.

Formula:
$P(A|B) = [P(B|A) \times P(A)] / P(B)$

Application:
Estimating the probability of promotion based on employee performance.

8. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique.

Advantages:
- Reduces the number of variables.
- Preserves maximum information.
- Removes redundancy.
- Improves visualization and model performance.

Applications:
Data analysis, machine learning, image processing and pattern recognition.

Conclusion

This report covers the theoretical concepts required for the practical examination. These concepts are applied in Python using the employee performance dataset to perform statistical analysis, probability calculations, data visualization and basic linear algebra.